Alvaro Gomariz

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Lead ML Scientist with over 10 years of experience designing, training, and adapting deep learning models to solve complex clinical problems. With a PhD in Computer Vision and 5+ years in the pharmaceutical industry, I specialize in foundation models, multi-modality, and uncertainty estimation. I am highly focused on rigorous statistical evaluation, ensuring model robustness, and translating cutting-edge research into scalable, high-impact health applications.

Experience

Roche, Switzerland — **Mar 2021 - Present**

Technical Lead - AI Video Colonoscopy — *May 2024 - Present*

- Technical and scientific leadership for AI video colonoscopy project, guiding a team of 10 ML scientists and engineers and orchestrating algorithm roadmaps aligned to prioritized use cases across cross-disciplinary stakeholders
- Design and execute rigorous model evaluation protocols beyond standard accuracy metrics, assessing model robustness, and performing subgroup analyses for clinical reliability
- Lead an annotation campaign with 5 renowned academic experts and 4 gastroenterologists to assess endoscopic scoring reliability and develop a new AI severity score
- Drive global engagement through conferences and internal forums, publishing results and filing a patent application, and establishing collaborations with global leaders in the field

Technical Lead - Foundation Models — *May 2023 - Apr 2024*

- Technically led a matrix team of 5–10 ML scientists, steering hands-on R&D for a company-wide multimodal foundation model initiative utilizing 35M+ digital pathology images, and another for colonoscopy videos
- Architected custom data engineering pipelines to handle gigapixel-scale inputs, optimizing distributed pre-training with robust data versioning for production
- Adapted state-of-the-art vision architectures (e.g., DINOv2) to deliver a proprietary model, resulting in a widely adopted internal software package and 3 lead-author papers
- Designed rigorous benchmarking frameworks to evaluate internal models against external vendors, directly guiding corporate build-vs-buy decisions
- Co-led an internal data science network of 25–35 experts, driving unified ML research methodologies and advanced engineering capabilities across the group

Data scientist — *Aug 2022 - Apr 2023*

- Built proof of concept on lung cancer patient stratification using ML on digital pathology images
- Steered and supervised the development of novel ML solutions (multiple instance learning, graph neural networks, self-supervised learning, survival analysis) and software development (MLops)
- Built group technical capabilities by identifying foundational scientific methodologies across different products and organizing matrix teams

Postdoctoral researcher — *Mar 2021 - Jul 2022*

- Conducted research to improve ophthalmic image segmentation algorithms, leading to improved predictions on images from new devices without annotated data
- Published methods with proposed methodologies (semi-supervised learning, contrastive CNNs, domain adaptation)
- Agile software development for integration of scientific outcomes into product development

Technical lead - MicroscopeIT, Poland & U.S.A. (remote) — **Mar 2018 - Mar 2019**

- Led R&D project from conception to delivery, creating an image analysis product leveraging deep learning and spatial statistics for large 3D microscopy clinical datasets
- Directed a team of 3-5 engineers to develop the software for biotechnology clients, including clinicians and researchers

Doctoral researcher - ETH & University Hospital Zurich, Switzerland — **Sept 2016 - Feb 2021**

- Conducted research on deep learning methods (attention networks, missing modalities, uncertainty estimation) to facilitate image analysis tasks (segmentation, detection, regression, and classification) of large 3D microscopy datasets, leading to multiple publications in renowned scientific journals
- Collaborated with biologists, clinicians, and healthcare companies, on multiple imaging domains (MRI, ultrasound, micro-CT, and different microscopy types), including software development for users with no programming experience, leading to scientific publications and research funding
- Wrote research grants resulting in project funding, supervision of 7 students, teaching, and organization of seminars

Education

Doctor of Science in Computer Vision - ETH Zurich, Switzerland **Oct 2016 - Feb 2021**

- Best PhD Award from the Center for Experimental and Clinical Imaging Technologies (EXCITE) Zurich for outstanding thesis in the field of biomedical imaging
- PhD thesis: "Quantitative Fluorescence Microscopy with Marker-Efficient and Uncertainty-Aware Segmentation and Detection Using Deep Learning"

MSc in Biomedical Engineering - ETH Zurich, Switzerland **Oct 2014 - Sep 2016**

- 1st of class: awarded Willi-Studer Prize
- Master thesis: "Applying 3D quantitative microscopy to study global topography and cellular interactions in the bone marrow"
- Semester project: "Investigation of image registration for sliding boundaries"

BSc in Biomedical Engineering - University Carlos III of Madrid, Spain **Oct 2010 - Sep 2014**

- 1st of class: Awarded Extraordinary End of Studies Award
- Bachelor thesis: "Resolution study of optical tomography systems in microscopy"

Skills

Machine learning	Foundation models, multi-modality, distributed pre-training, GNNs, uncertainty estimation
Software development	MLOps, data engineering, CI/CD, ETL, high performance computing (SLURM & LSF), git
Programming & Scripting	Python, SQL, Matlab, bash/shell, Docker, R
Frameworks & tools	PyTorch, TensorFlow, AWS, V7, Nextflow, SciPy Ecosystem, ITK, OpenCV
Project leadership	Matrix teams, strategic planning, product development, cross-functional collaboration
Agile certifications	ICAgile Certified Professional (May 2024), Professional Scrum Master^{TM} I (Aug 2022)
Languages	English (fluent), Spanish (native), German (basic)

Additional information

Scientific Contributions

Journal articles

- **Gomariz A**, Kikuchi Y, Li Y, Albrecht A, Maunz A, Ferrara D, Lu H, Goksel O, "Joint semi-supervised and contrastive learning enables domain generalization and multi-domain segmentation", *Medical Image Analysis* 103, p.103575 (2025). doi: 10.1016/j.media.2025.103575
- Yamada Y, Nguyen TT, Impellizzieri D, Mineura D, Shibuya R, **Gomariz A**, Haberecker M, Nilsson J, Nombela-Arrieta C, Jungraithmayr W, Boyman O "Biased IL-2 signals induce Foxp3-rich pulmonary lymphoid structures and facilitate long-term lung allograft acceptance in mice", *Nature Communications* 1383 (14) (2023). doi: 10.1038/s41467-023-36924-z
- Vijaykumar A, Helbling PM, Thelen F, Fazio S, Mun Y, Pereira AL, Zielińska KA, Buschl P, Zerkatje T, Loosli K, Isringhausen S, Menze B, Nagasawa T, Roeder I, **Gomariz A**, Manz MG, Yokomizo T, Nombela-Arrieta C "Tissue-scale mapping reveals a central role of hepatoblasts in the regulation of fetal liver hematopoiesis and stem cell maintenance", *Developmental Cell* 61, 901-918.e4 (2026). doi: 10.1016/j.devcel.2026.01.011
- **Gomariz A**, Portenier T, Nombela-Arrieta C, Goksel O, "Probabilistic Spatial Analysis in Quantitative Microscopy with Uncertainty-Aware Cell Detection using Deep Bayesian Regression", *Science Advances* 8 (5) (2022). doi: 10.1126/sciadv.abi8295
- **Gomariz A**, Portenier T, Helbling PM, Isringhausen S, Suessbier U, Nombela-Arrieta C, Goksel O, "Modality Attention and Sampling Enables Deep Learning with Heterogeneous Marker Combinations in Fluorescence Microscopy", *Nature Machine Intelligence* p. 1-13 (2021). doi: 10.1038/s42256-021-00379-y
- Isringhausen S, Mun Y, Kovtonyuk L, Krautler N, Suessbier U, **Gomariz A**, Spaltro G, Helbling PM, Wong HC, Nagasawa T, Manz MG, Oxenius A, Nombela-Arrieta C, "Chronic viral infections persistently alter marrow stroma and impair hematopoietic stem cell fitness ", *Journal of Experimental Medicine* 218 (12) (2021). doi: 10.1084/jem.20192070
- Ghazi S, Bourgeois S, **Gomariz A**, Bugarski M, Haenni D, Martins JR, Nombela-Arrieta C, Wagner CA, Hall AM, Eilidh C, "Multiparametric imaging reveals that mitochondria rich intercalated cells in the kidney collecting duct have a very high glycolytic capacity", *The FASEB journal* 34 (6), pp. 8510–8525 (2020). doi: 10.1096/fj.202000273R
- **Gomariz A**, Isringhausen S, Helbling PM, and Nombela-Arrieta C, "Imaging and spatial analysis of hematopoietic stem cell niches", *Annals of the New York Academy of Sciences* 1466(1):5-16 (2019). doi: 10.1111/nyas.14184
- Hess M, **Gomariz A**, Goksel O, Ewald CY, "In-vivo quantitative image analysis of age-related morphological changes of C. elegans neurons reveals a correlation between neurite bending and novel neurite outgrowths", *eNeuro* 6(4) (2019). doi: 10.1523/ENEURO.0014-19.2019
- Agarwal P, Isringhausen S, Li H, Paterson AJ, He J, **Gomariz A**, Nagasawa T, Nombela-Arrieta C, and Bhatia R, "Mesenchymal niche-specific expression of Cxcl12 controls quiescence of treatment-resistant leukemia stem cells", *Cell Stem Cell* 24 (5), 769–784.e6 (2019). doi: 10.1016/j.stem.2019.02.018
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Becker A, Boss A, Nagasawa T, Paul G, Goksel O, Székely G, Stoma S, Nørrelykke SF, Manz MG, and Nombela-Arrieta C, "Quantitative spatial analysis of haematopoiesis- regulating stromal cells in the bone marrow microenvironment by 3D microscopy", *Nature Communications* 9 (1) (2018). doi: 10.1038/s41467-018-04770-z
- Takizawa H, Fritsch K, Kovtonyuk LV, Saito Y, Yakkala C, Jacobs K, Ahuja AK, Lopes M, Hausmann A, Hardt WD, **Gomariz A**, Nombela-Arrieta C, and Manz MG, "Pathogen-Induced TLR4-TRIF Innate Immune Signaling in Hematopoietic Stem Cells Promotes Proliferation but Reduces Competitive Fitness", *Cell Stem Cell* 21 (2), 225–240.e5 (2017). doi: 10.1016/j.stem.2017.06.013

Conference papers

- Blasco Fernandez P, Bossuyt P, Daperno M, Kopylov U, Bouhnik Y, Karmiris K, Benmansour F, Gutierrez Becker B, Fraessle S, Stimpel B, Levitte S, **Gomariz A** "Quantifying Inter-Rater Reliability of Mucosal Features in Ulcerative Colitis Endoscopy", *Journal of Crohn's and Colitis* Volume 20, Issue Supplement_1 (2026). doi: 10.1093/ecco-jcc/jjaf231.052
- **Gomariz A**, Gutierrez-Becker B, Girycki R, Szostak P, Stimpel B, Levitte S, Abbasi-Sureshjani S "Assessing the Impact of Biopsy Artifacts on AI-Based Endoscopic Scoring for Ulcerative Colitis", *Journal of Crohn's and Colitis* Volume 20, Issue Supplement_1 (2026). doi: 10.1093/ecco-jcc/jjaf231.699
- Albuquerque T, Yuce A, Herrmann MD, and **Gomariz A**, "Characterizing the Interpretability of Attention Maps in Digital Pathology", *arXiv* (2024). doi: 10.48550/arXiv.2407.02484
- Bredell G, Fischer M, Szostak P, Abbasi-Sureshjani S, and **Gomariz A**, "The Importance of Downstream Networks in Digital Pathology Foundation Models", *International Workshop on Foundation Models for General Medical AI - MICCAI* pp. 10-19 (2024). doi: 10.1007/978-3-031-73471-7_2
- Ibañez v, Szostak P, Wong Q, Korski K, Abbasi-Sureshjani S, and **Gomariz A**, "Integrating multiscale topology in digital pathology with pyramidal graph convolutional networks", *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 1-4 (2024). doi: 10.1109/ISBI56570.2024.10635702
- Kikuchi Y, **Gomariz A**, Li Y, Lu H, Albrecht A, Ferrara D, "Device adaptation of optical coherence tomography (OCT) retinal layer segmentation algorithm using unlabeled target data", *Investigative Ophthalmology & Visual Science, ARVO abstract (2023)*
- **Gomariz A**, Lu H, Li Y, Albrecht A, Maunz A, Benmansour F, Valcarcel AM, Luu J, Ferrara D, Goksel O, "Unsupervised Domain Adaptation with Contrastive Learning for OCT Segmentation", *International Conference on Medical image computing and computer-assisted intervention (MICCAI)* pp. 351-361 (2022). doi: 10.1007/978-3-031-16452-1_34
- Li Y, Lu H, Albrecht T, Maunz A, Neubert A, Benmansour F, **Gomariz A**, Luu J, Ferrara D "Deep learning (DL) to segment retinal layer disruption on optical coherence tomography (OCT) in neovascular age-related macular degeneration (nAMD)", *Investigative Ophthalmology & Visual Science, ARVO abstract (2022)*
- **Gomariz A**, Lu H, Li Y, Albrecht A, Maunz A, Benmansour F, Luu J, Goksel O, Ferrara D, "A unified deep learning approach for OCT segmentation from different devices and retinal diseases", *Investigative Ophthalmology & Visual Science, ARVO abstract (2022)*
- **Gomariz A**, Egli R, Portenier T, Nombela-Arrieta C, and Goksel O, "Utilizing Uncertainty Estimation in Deep Learning Segmentation of Fluorescence Microscopy Images with Missing Markers", *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 371-374 (2021). doi: 10.1109/ISBI48211.2021.9434158
- **Gomariz A**, Li W, Ozkan E, Tanner C, and Goksel O, "Siamese Networks with Location Prior for Landmark Tracking in Liver Ultrasound Sequences", *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 1757-1760 (2019). doi: 10.1109/ISBI.2019.8759382
- Goksel O, Vishnevsky V, **Gomariz A**, and Tanner C, "Imaging of Sliding Visceral Interfaces During Breathing", *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 298-301 (2016). doi: 10.1109/ISBI.2016.7493268